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## Exploring the link between inhibitory control and cognitive flexibility in preschool children

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### **Abstract**

*The preschool years are considered the most important period in terms of executive functions development. In this study we explored the link between two executive functions in preschool children: inhibitory control and cognitive flexibility. For the assessment of inhibitory control, we used the Multiple Choice Reaction Test and as the measure of cognitive flexibility we used Semantic Fluency Test. The sample for this study consisted of 72 preschool children (35 girls;  $M_{age} = 59.2$  months,  $SD = 9.2$  months). The results of this study indicated (1) an overall low correlation between inhibitory control and cognitive flexibility (yet nonsignificant in boys, and moderate correlation in girls); (2) girls outperformed boys on the inhibitory control task, but not on the cognitive flexibility one; (3) children's age was significantly correlated with their cognitive flexibility performance, but not with their inhibitory control scores. Our overall findings indicate that inhibitory control and cognitive flexibility are distinct features of executive functioning and seem to have different trajectories in preschool children as a function of gender and age.*

**Keywords:** preschool children, executive functions, inhibitory control, cognitive flexibility

The preschool years represent the most important period for the development of executive functions (EFs), a crucial predictor of children's later academic outcomes, and of their later success in life (Diamond, Barnett, Thomas, & Munro, 2007). Although the researchers are not in agreement about what dimensions are part of EF, most of them agree that EF consists of Inhibitory Control (self-control, self-regulation), Working memory and Cognitive flexibility (Diamond & Lee, 2011; Ng, Tamis-LeMonda, Yoshikawa, & Sze, 2015).

Inhibitory control can be defined as the ability to control thought processes and actions that can potentially interfere with the individual's goal (Carlson &

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Wang, 2007). In addition to this, inhibitory control plays an important role in determining how different cognitive processes work together in order to successfully accomplish a task (Dowsett & Livesey, 2000). The development of inhibitory control is particularly rapid in preschool years and the ability to suppress irrelevant stimuli and actions significantly improves with age (Durstun et al., 2002). In this respect, existing studies reveal that inhibitory control abilities go through a drastic change in the 3<sup>rd</sup> year of life (Posner & Rothbart, 2000). Most children at that age have difficulties on the tasks requiring inhibitory control of attention and motor response which significantly improves by the age of 5 years (Carlson & Wang, 2007). Inhibitory control is also regarded as part of a self-regulation (Ponitz, McClelland, Matthews, & Morrison, 2009). According to the views of Posner and Rothbart (2000), the understanding of self-regulation is one of the most important requirements for understanding development and psychopathology. Indeed, research has demonstrated that inhibitory control is associated with children's adjustment, problems in externalized behavior, distractibility and impulsivity (Logan, Schachar, & Tannock, 1997; Rotenberg, Michalik, Eisenberg, & Betts, 2008; Schachar & Logan, 1990). In relation to the effects of gender on inhibitory control, research has shown that girls have higher inhibitory control than boys (Else-Quest, Hyde, Goldsmith, & Van, 2006). In this study we used number of Commission Errors on Multiple Choice Reaction Test as an indicator of inhibitory control.

Another executive function of great importance to the child's overall cognitive and affective functioning is cognitive flexibility. This is regarded as a complex process involving many behaviors that are considered flexible such as multitasking, novelty generation, or flexible problem solving, thus making the concept of cognitive flexibility not an easy one to define (Ionescu, 2012). One of the definitions used refers to cognitive flexibility as an ability to easily shift attentional focus in order to perceive, process and respond to situations in different ways (Eslinger & Grattan, 1993). Another aspect of cognitive flexibility, as the name implies, is the flexibility of ideas and the adjustment to change. Through this conceptualization, it is evident that cognitive flexibility is strongly related to creativity (Dreu, Nijstad, & Baas, 2011) and its assessment at preschool age is of crucial importance. Cognitive flexibility is involved in the acquisition of many skills that occur in the childhood such as language, math skills, theory of mind and interpersonal interactions (Chevalier & Blaye, 2009). In addition to this, better cognitive flexibility at preschool age is related to better school readiness (Vitiello, Greenfield, Munis, & George, 2011). Assessment of cognitive flexibility mainly depends upon which segment of flexibility we want to assess such as set-shifting, flexible categorization, flexible language use etc. In preschool years is assessment is usually performed with card sorting tasks such as Dimensional Change Card Sort in which children must sort cards based on their specific perceptual dimension (Low & Simpson, 2012). Another way to assess cognitive flexibility is through the verbal

fluency tasks (Koren, Kofman, & Berger, 2005; Paula, Paiva, & Costa, 2015). There are two types of verbal fluency tests, phonological/orthographic fluency tests and semantic fluency tests. For younger children it is more appropriate to use semantic fluency tests than orthographic fluency tests (Welsh, Pennington, & Groisser, 1991). Thus, in this study we assessed the construct of cognitive flexibility through the semantic fluency task. Unlike Card Sorting testing paradigm, which mainly have dichotomous outcome, the semantic fluency provides dimensional measure of performance and thus is more informative for the subsequent statistical analysis. It is important to mention that semantic fluency test is not a pure measure of cognitive flexibility as it also taps the domain of general verbal functioning (Memisevic, Ibralic-Biscevic, & Pasalic, 2017a). Language skills such as labeling, especially at preschool age, play an important role in cognitive flexibility tasks (Low & Simpson, 2012). However, there are few, if any, “pure” measures of EF domains and most of the measures also tap some other cognitive skills such as working memory (Carlson, 2005). Moreover, some perspectives postulate that the constructs of cognitive flexibility and inhibitory control strongly intertwined and interdependent (Chevalier & Blaye, 2008).

With regards to potential gender differences, studies examining cognitive flexibility through verbal fluency tasks have found mixed results, with some studies suggesting that success on this task strongly depends on the semantic category used (Memisevic et al., 2017a).

*Current study.* Given the importance of inhibitory control and cognitive flexibility, relatively little research has examined the relationship between the two EF constructs in preschool children. Examination of the nature of this relationship is especially relevant to determine whether the two constructs are separate at preschool age. Existing studies have focused on the relation of these constructs with theory of mind (Low & Simpson, 2012) or on the effects of preschool programs on these domains (Diamond & Lee, 2011) but less on the nature of the link between inhibitory control and cognitive flexibility. Aside from examining the relationship of these two constructs at preschool age, we also aimed to examine how the two EF domains and their association vary as a function of children’s age and gender.

Thus, the first goal of the study was to examine the link between inhibitory control and cognitive flexibility in preschool children. In this respect, given the interdependence indicated by previous research, we expected a positive correlation between inhibitory control and cognitive flexibility. Second, we aimed to examine inhibitory control and cognitive flexibility variations as a function of children’s gender. Based on previous findings, we expected girls to score higher on the inhibitory control task, yet no significant differences on the cognitive flexibility task engaged in this study. Finally, we also wanted to determine how children’s performance on the two EF tasks varies as a function of their age. In this respect, based on previous findings, we expected older children to display superior

performances on both tasks, denoting a significant age-related EF improvement during the preschool years.

## METHOD

### Participants

The sample for this study consisted of 72 preschool children aged 3 to 6 years ( $M_{\text{age}} = 59.2$  months,  $SD = 9.2$  months). There were 35 girls (48.6%) and 37 boys (51.4%) in the sample. There were no statistically significant differences in the mean age between boys and girls ( $t = 0.48$ ;  $p = .63$ ). Children were attending public preschool institutions in Canton Sarajevo, Bosnia and Herzegovina. They were all free of any known neurological and/or psychiatric conditions or developmental disabilities.

### Procedure

Out of 30 public kindergartens in Canton Sarajevo, we selected 8 small-sized public kindergartens (up to 50 children) and provided preschool teachers with the consent forms for children's parents. The goals of the study were explained to the preschool teachers. Consent forms contained all the information regarding the study. It was pointed out that participating in the study is on voluntary basis and that the obtained data will be analyzed anonymously. This study was part of a larger study examining executive function of preschool children in Canton Sarajevo. After the consent forms were returned (out of 200 consent forms, 158 were signed and returned, 79% response rate), we tested the children on a number of tests. Children were tested by trained data collectors with advanced degrees in psychology, speech and language therapy and special education. All children were tested individually, in the morning hours, in the convenient space at the kindergartens. The complete testing session lasted approximately 20 to 25 minutes for each child. Some children were sick at the time of testing ( $n=15$ ) or did not want to do the tests ( $n=10$ ), or the testing data were incomplete ( $n=17$ ). We also excluded data of children where the reaction time on Multiple Choice Reaction Test was less than 0.3s as it might indicate accidental responding. Data from two children were excluded as they represented significant outliers. This left us with the final sample of 72 children from these 8 kindergartens for whom we had the complete data. Out The approval for this study was obtained from the Canton Sarajevo Ministry of Education and the Ethical Committee Board at the Faculty of Educational Sciences at the University of Sarajevo. Only children with the written parental consent were tested.

## Measures

*Inhibitory control.* The Multiple-Choice Reaction Time Test is a computerized test in which children are required to press the space button on a computer any time they see a star appearing on the screen (Di Nuovo, 2000). Children needed to inhibit the response (pressing a space button) when they see other objects appearing on the screen (go-no/go paradigm). The total number of stimuli presented was 45, out of which 9 are targets. Children were requested to press a space button each time the target appears on the screen. All stimuli were presented randomly to children. The program used for this assessment was *Attenzione e concentrazione* (Di Nuovo, 2000). There are four outcome measures on this test: number of correct answers, mean reaction time, errors of omission and errors of commission. For the purposes of this study we only present the number of commission errors committed as a measure of inhibitory control. Commission errors are seen as a measure of inhibitory control. The other three measures, number of correct answers, mean reaction time, and number of omission errors are regarded as the measures of attentional control and were not used as a dependent variable in this study.

*Cognitive flexibility.* For the purposes of this study we used a Semantic Fluency task, with a semantic category of animals as this is the most widely used category in semantic fluency tasks. This task, especially in preschool children, measures their ability to cluster and switch between the different subcategories- or in other words measures their cognitive flexibility. Children who are able to cluster more items and switch between the categories faster (e.g. sea animals, forest animals, etc.) will name more items. In this task children were asked to name as many animals as they could in 60seconds. All children understood the instructions of the task and were able to perform it. The total number of animals (non-repeated) retrieved was used as a dependent measure.

## Analytic approach

For the first goal of this study, the relationship between inhibitory control and cognitive flexibility, we calculated the Pearson correlation coefficient for the whole sample, followed by calculating this for boys and girls separately. For the second goal of the study, we analyzed the statistical differences in mean performance scores for boys and girls using the t-test. As a measure of an effect size, we calculated Cohen's *d* coefficient. Finally, we calculated intercorrelations between age, inhibitory control and semantic fluency for the whole sample, followed by the intercorrelations as a function of the child's gender. Data were analyzed with the computer program SPSS for Windows (v.13).

## RESULTS

The first goal of this study was to examine the relationship between children's inhibitory control and cognitive flexibility scores. The results showed that there is small to medium relationship between the two constructs (Pearson  $r = -.26$ ;  $p = .03$ ). However, gender of the child had a strong moderating effect on this correlation. For the girls the correlation between inhibitory control and cognitive flexibility was much stronger (Pearson  $r = -.41$ ;  $p = .016$ ) than in boys (Pearson  $r = -.12$ ;  $p = .49$ ).

The second objective in this study was to examine the differences in inhibitory control and cognitive flexibility in relation to the child's gender. The results are shown in Table 1.

Table 1.

*Mean scores of boys and girls on inhibitory control and cognitive flexibility tasks*

| Executive function task | Boys |      | Girls |      | t (70) |
|-------------------------|------|------|-------|------|--------|
|                         | M    | SD   | M     | SD   |        |
| Inhibitory control      | 1.62 | 1.45 | 0.91  | 1.10 | -2.34* |
| Cognitive flexibility   | 6.76 | 3.47 | 7.83  | 2.95 | 1.41   |

Note: \* $p = .022$ ; the inhibitory control measure entitles measuring errors and not accuracy.

As it can be seen from Table 1, girls had fewer commission errors than boys on the task of inhibitory control and this difference was statistically significant. The effect of this difference, as measured by Cohen's  $d$ , was medium to large (Cohen's  $d = 0.55$ ). Although the girls achieved higher mean score on the task of cognitive flexibility, this difference was not statistically significant ( $p = .16$ ) and the Cohen's  $d$  effect size was small to medium (Cohen's  $d = 0.33$ ).

The third objective in this study was to examine the influence of age on inhibitory control and cognitive flexibility. Age was significantly related with cognitive flexibility but not with inhibitory control. These correlations are shown in Table 2.

Table 2.

*Correlations between age, inhibitory control and cognitive flexibility*

| Variable                 | 1     | 2     | 3 |
|--------------------------|-------|-------|---|
| 1. Age                   | –     |       |   |
| 2. Inhibitory control    | -.02  | –     |   |
| 3. Cognitive flexibility | .38** | -.26* | – |

Note. \* $p < .05$ ; \*\* $p < .01$ .

We further examined whether these correlation trends were similar among boys and girls. The correlations as a function of children's gender are shown in Table 3.

Table 3.

*Correlations between age, inhibitory control and cognitive flexibility as a function of children's gender*

| Variable                 | 1     | 2    | 3     |
|--------------------------|-------|------|-------|
| 1. Age                   | –     | -.14 | .34*  |
| 2. Inhibitory control    | .01   | –    | .41** |
| 3. Cognitive flexibility | .47** | .12  | –     |

*Note.* \* $p < .05$ ;  $p < .01$ . Correlations for the girls ( $n = 35$ ) are presented above the diagonal (italic), and the correlations for the boys ( $n = 37$ ) are presented below the diagonal.

Analyses indicate that the relationship between inhibitory controls and age was small and non-significant for both boys and girls (see Table 3). Age was significantly correlated with cognitive flexibility in boys and girls, although this relationship seems relatively stronger in boys. Finally, inhibitory control was significantly correlated with cognitive flexibility only in girls but not in boys.

## DISCUSSIONS

The main aim of the present study was to examine the relationship between two EF components, inhibitory control and cognitive flexibility. The results of this study revealed small correlation between the two constructs, thus providing more evidence that the two constructs are distinct in preschool children. However, we have also found that the relationship between the two constructs varied as a function of children's gender. For girls, this relationship was medium in size and statistically significant, while for boys the relationship was statistically nonsignificant. Possible explanations for the differential correlation as a function of gender are related to gender differences in the functional organization of the brain for language, as well as the differences in terms of information processing speed. Studies have shown that brain activity in language tasks is more diffuse in females, involving both cerebral hemispheres, while in males it is lateralized in the left hemisphere frontal regions (Shaywitz, Shaywitz, Pugh, & Constable, 1995). It is also indicated that males and females are using different processing strategies in verbal fluency tasks (Weiss et al., 2006). Another possible explanation might be the confounding effect of other executive functions constructs such as working memory. In this respect, there is convincing evidence the brain structures activated during a performance on working memory tasks differ between males and females (Speck et al., 2000). Of course, there is always a possibility that different developmental trends observed were sample-dependent and are not representative of the population of preschool children, especially given the relatively small sample size.

We further examined gender differences on the inhibitory control task. The boys had poorer performances as they committed more errors on this task than the girls.

This finding is in line with existing studies showing that girls were better at inhibitory control than boys (Kochanska, Murray, Jacques, Koenig, & Vandegeest, 1996).

Deficits in inhibitory control are particularly remarkable in certain neurodevelopmental disorders such as ADHD and autism (Christ, Holt, White, & Green 2007; Schachar, Mota, Logan, Tannock, & Klim, 2000). As inhibitory control is partially susceptible to training, improving it would certainly lead to amelioration of some symptoms and lead to better academic and adaptive outcomes of children with inhibitory control deficits. It is important to note that not all EFs are equally susceptible to training effects. This might be due to the fact that different EFs have different neural correlates. For example, it seems to be more difficult to train inhibitory control compared to training working memory (Thorell, Lindqvist, Bergman Nutley, Bohlin, & Klingberg, 2009).

Another finding in this study was that boys and girls did not statistically differ in cognitive flexibility as measured through the task of semantic fluency. There are inconclusive results on whether there are some inherent differences in semantic fluency in relation to gender (Capitani, Laiacona, & Barbarotto, 1999; Memisevic et al., 2017a), yet it might be that efficacy on the semantic fluency task depends on the category used and the exposure of the child to the particular category.

We have also examined the effects of age on inhibitory control and cognitive flexibility. In line with previous studies on school-aged children (Riva, Nichellim, & Devoti, 2000), age was positively associated with cognitive flexibility in both boys and girls. This is particularly important as preschoolers' increasing age is related to the improvement in many other areas such as motor control and speed (Memisevic, Mahmutovic, Pasalic, & Bisevic, 2017b), and improvements in language abilities (Pullen & Justice, 2003). However, while the preschool years are the most important for the development of cognitive flexibility, depending on the measure used, this EF continues to substantially improve until adolescence, with the greatest improvements being between 6 and 10 years of age (Cepeda, Kramer, & Gonzalez de Sather, 2001). This is valid for Wisconsin Card Sorting Test as well as for the semantic fluency test.

*Limitations.* First, as the sample was relatively small, we must be cautious in generalizing these results. Secondly, we assessed domains of inhibitory control and cognitive flexibility with only one test per each domain. Future studies should focus on more comprehensive measures of both domains. Finally, it would be useful in this analysis if we could control for the effects of IQ and connected EFs, such as working memory.



## CONCLUSIONS

Inhibitory control and cognitive flexibility are distinct executive functions at preschool age. The correlation between these two constructs was small to medium. The results of this study indicated (1) an overall low correlation between inhibitory control and cognitive flexibility (yet nonsignificant in boys, and moderate correlation in girls); (2) girls outperformed boys on the inhibitory control task, but not on the cognitive flexibility one; (3) children's age was significantly correlated with their cognitive flexibility performance, but not with their inhibitory control scores. Findings of this study can inform researchers aiming to develop EF training programs for young children about the specificities of inhibitory control and generative fluency at this age. In this respect, our overall findings indicate that inhibitory control and cognitive flexibility are distinct features of EF and seem to have different trajectories in preschool children as a function of gender and age. For some executive functions there is already a large body of evidence demonstrating the possibility of its improvement. Probably most attention has been given to working memory training and the evidence for the potential of its improvement is relatively strong (Klingberg, 2010; Morrison & Chain, 2011). Much less attention has been given to the training of inhibitory control and cognitive flexibility. However, there are some promising studies that examined interventions aimed at improvement inhibitory control. For example, early music education has been shown to improve inhibitory control (Joret, Germeys, & Gidron, 2017). Early interventionists, psychologists and educators need to be aware of these programs and potentially incorporate them into the preschool curricula.

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